

Yunzhe Zhang

MACHINE LEARNING RESEARCHER

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Research Area

Interested in Generative AI for real-world applications: building efficient, effective, and explainable models; current focus on controllable diffusion and multimodal learning, with emphasis on improving inference and efficiency strategies for stable, steerable text-to-image synthesis.

Education

University of Illinois at Urbana Champaign

B.S. IN COMPUTER SCIENCE AND STATISTICS

- GPA: 3.83 / 4.0.
- Major GPA: 3.91 / 4.0.

Urbana, IL, USA

Aug. 2018 - May. 2021

Columbia University

M.S. IN DATA SCIENCE

- GPA: 3.70 / 4.0.

New York, NY, USA

Aug. 2021 - Dec. 2022

Brandeis University

P.H.D. IN COMPUTER SCIENCE ADVISED BY PROFESSOR PENGYU HONG

- GPA: 4.0 / 4.0.

Waltham, MA, USA

Sep. 2023 - May. 2028

Research Experience

Boosting Text-to-Image diffusion models via attention-driven seed selection

FIRST AUTHOR IN PROGRESSING SUBMIT TO ICLR 2026

- Advisor: Professor Hongfu Liu and Professor Pengyu Hong
- Proposed ADSS (Attention-Driven Seed Selection), a training-free, plug-and-play scheme that ranks seeds via early cross-attention concentration on prompt core/body tokens. The lightweight scoring enables per-prompt seed preselection without fine-tuning or latent changes and is architecture-agnostic.
- Improved image relevance and quality across SD 1.4/1.5/2.0/2.1 and three public benchmarks (DrawBench, InitNO, Pick-a-Pic), outperforming other state-of-the-art seed selection methods across four metrics simultaneously; for example, on DrawBench with SD 2.1, IR increased by 0.0409 and HPS increased by 0.0025. Unlike prior methods that require substantial computation to curate and evaluate large validation sets, ADSS requires only a lightweight screening stage; per-seed screening takes about 2.72 seconds.
- Delivered practical significance by enabling per-prompt seed preselection and early stopping that improve sample efficiency and reliability; ADSS also serves as an add-on to existing optimization methods (e.g., seed/noise optimization) to further boost performance without any re-training or architectural changes.

Waltham, USA

Feb. 2025 - Aug. 2025

Cross-Modality Controlled Molecule Generation with Diffusion Language Model

FIRST AUTHOR IN PROGRESSING SUBMIT TO AAAI 2026

- Paper: <https://arxiv.org/pdf/2508.14748>.
- Advisor: Professor Pengyu Hong
- Proposed the idea of controllable SMILES generation under joint property-structure objectives, introducing a scalable diffusion-based paradigm that composes multiple property and scaffold constraints, boosting target-property satisfaction by 16% across three benchmarks and achieving average 55% scaffold similarity.
- Implemented classifier-guided conditional generation by training multiple property predictors and injecting their gradients into the reverse diffusion process, enabling targeted control of QED, SAS, and (P)LogP.
- Co-developed classifier-free, structure-aware conditioning: froze the pretrained diffusion backbone and integrated condition embeddings plus cross-attention into the Transformer to improve scaffold controllability while preserving validity and diversity.
- Designed a two-phase inference scheme: early steps anchor the molecular scaffold via a Structure Control Module, followed by a Property Control Module with score fusion to balance structure fidelity and property optimization; the modules are plug-and-play and composable.

Waltham, USA

Feb. 2024 - Jan. 2025

Reinforcement Learning Enabled Energy Optimization For Prosumers

New York, USA

TEAM LEADER AND RESEARCH ASSISTANT

Sep. 2022 - Dec. 2022

- Advisor: Dr. Bhushan Jagyasi (Associate Director Data Science at Accenture)
- Focused on developing a reinforcement learning based agent for prosumers and consumers to minimize the overall energy cost and increase the renewable energy utilization
- Conducted the Exploratory Data Analysis and proposed key ideas to address the optimization in terms of the MDP formulation including states, control actions, and rewards
- Built the environment simulator involving rest, render, and step function based on the previous MDP formulation for different appliances including Shiftable Appliances with or without Interruptible Load, Reducible Appliances, and ESS
- Collaborated with other teammates to do the Agent Training including tabular Q-learning, Deep Q-learning, and PPO2 by using the best parameters resulted from Tensorboard automated hyper-parameter tuning and found the agent using DQN performs the best

NLP approach to EPA text analytics classification and scoring

New York, USA

RESEARCH ASSISTANT

Sep. 2021 - May. 2022

- Advisor: Adjunct Professor Anhphan Ly
- Worked on leveraging machine learning and natural language processing techniques to optimize the way in which evaluate how well students are hitting EPA standards so that administrators and students could sort and classify their qualitative feedback
- Refined the code finished by previous intern students and trained the models including XGBoost, and CNN on the new labelling data to check and compared the results with the previous attempt results
- Collaborated with other master and Ph.D. students to develop the BLSTM with 0.5 dropout rate and 380k parameters, and BERT model with 0.1 dropout rate and 0.25 for classification and sentiment analysis tasks to regularize and prevent over-fitting
- Investigated relation between different models and its performance for each EPA levels; concluded XGBoost have relatively higher precision, recall, and F1 score for "Knowledge" and BLSTM achieves the highest 0.802 macro averaged F1 score to be the best model for EPA classification

Projects

AD Click Fraud Detection Project

New York, USA

DATA ANALYST AND MACHINE LEARNING MODEL DEVELOPER

Sep. 2021 - Dec. 2021

- Worked on AD Click Fraud Detection project which is used to predict whether an app will be downloaded after its advertisement is clicked
- Contributed to the Exploratory Data Analysis and solved insufficient memory usage, lacking of descriptive features, and highly imbalanced dataset problems
- Implemented the pipelines and built the machine learning models including Support Vector Machine, Random Forest, XGBoost, and LightBGM. Finished plotting the evaluation matrix graph and POC-Curve
- Concluded that XGBoost being the best model proved by recall, average precision and F1 score, with a drawback of significantly longer training times; with limited resources, LightBGM would be chosen for deployment

FlightOrTrain Price Search Engine with Smart Recommendation

Urbana, USA

FULL STACK DEVELOPER

July. 2020 - Aug. 2020

- Designed and developed a website for flight price search, includes user interface, database and recommendation
- Implemented website UI based on React framework and improved user experience through conducting surveys
- Decided database entity diagram and realized it with insertion, deletion and update function using Python
- Realized flight recommendation based on the selected dates, available price and number of stops with the support of sorting on different attributes

Work Experience

Wuxi AppTec. Department of Digital Innovation

Shanghai, China

DATA SCIENTIST INTERN

Jun. 2020 - Sep. 2020

- Participated in the development of multi-language translation App and contributed to the EDA and data preprocessing part
- Tested Tencent, YouDao, KeDaXunFei, and BaiDu OCR precision rate via API testing and decided to utilize YouDao for best performance of 98% recognition rate
- Improved the OCR recognition precision by different levels of image segmentation and making using of both local and global information for OCR conversion
- Parsed the OCR results to form unified hierarchical data using Python and output in excel format as further training data
- Served as training data for later NLP model to better recognize term alias and found 15% more synonyms correctly

EDU-ESIGN Start-Up Company. Department of System development

Beijing, China

DATA SCIENTIST INTERN

Nov. 2020 - Feb. 2021

- Participated in the development of the recommendation system of the App and decide to custom questionnaire for new users to avoid large cost of using other company's data
- Conducted literature review on popular recommendation system algorithms, including content-based recommendation, collaborative filtering and model-based algorithms
- Trained SVM model and utilized Factorization Machine and Association Retrieval to solve feature sparsity resulted by one-hot encoding
- Improved the product cold-start problem by recommending trending content and making using of user questionnaire results, which achieves increasing the number of new online users by 20 %

Honors & Awards

2018-2021 **Dean's List**, Grainger College of Engineering and College of Liberal Arts and Sciences, UIUC

Champaign, IL

2018-2021 **Highest Graduation Distinct**, Grainger College of Engineering and College of Liberal Arts and Sciences, UIUC

Champaign, IL

Relevant Coursework

Machine Learning

COMS 4995(Applied Machine Learning), COMS 4705 (Natural Language Processing), COMS 4721 (Machine Learning for Data Science), COMS 4774 (Unsupervised Learning), CS 498 (Applied Machine Learning), MATH 415 (Linear Algebra)

Stats

STAT 5701 (Probability for Data Science), STAT 5702 (Exploratory Data Analysis), STAT 5703 (Statistics inference and modeling), STAT 428 (Statistical Computing), STAT 400 (Statistics and Probability I), STAT 410 (Statistics and Probability II), STAT 420 (Method of Applied Statistics)

Computer Science

COMS 4246 (Algorithm for Data Science), COMS 4121 (Computer System for Data Science), CS 421 (Programming Languages and Compilers), CS 411 (Computer Database), CS 374 (Introduction to Algorithms and Models of Computation), CS 357 (Numerical Method), CS 241 (System Programming), CS 233 (Computer Architecture), CS 225 (Data Sturcture), CS 173 (Discrete Structure)

Skills

Language

English, Mandarin

Programming

Python, C++, C, Java, JavaScript(React, NodeJS), CSS, HTML, SQL, R, MongoDB, Haskell, Latex

Machine Learning

Unsupervised Learning (Clustering, Dimension Reduction), Deep Learning (Natural Langhuage Processing, Graph Neural Netwrok), Reinforcement Learning (TD-learning, Deep Q-learning, Markov Decision Process)